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Characterising preferential flow and its interaction with the soil matrix using dye tracing in the Three Gorges Reservoir Area of China

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Abstract. Dye tracing experiments provide direct visual evidence of preferential flow in the soil. In this study, we applied the Brilliant Blue tracer across three forest sites (high-mountain forest, HF; middle-mountain forest, MF; and low-mountain forest, LF) and one cultivated field (CL) in the Three Gorges Reservoir Area of China to visualise preferential flow and characterise its interaction with the surrounding soil matrix. A set of parameters was extracted from photographs of dye-stained soil profiles to measure preferential flow, including (1) the ratio of the stained area to the total area of a soil section (SAR), (2) the degree of lateral water mixing of preferential flow into the soil matrix (LWM), (3) the greatest stained depth (SD), and (4) the stained path width (SPW). The highest SAR of all of the stained areas (i.e. a measure of the degree of preferential flow) was for MF (80%), followed by LF (68%), CL (48%), and HF (30%). The higher SAR in MF and LF was likely associated with more abundant and interconnected void spaces created by roots and soil fissures. The shallower rooting depth together with the higher content of clay and soil organic matter might lead to the lowest SAR in HF, suggesting a higher likelihood of soil erosion due to surface runoff. The relatively lower SAR in CL could be a result of soil compaction after tillage destroyed soil macropores. Moreover, the spatial distribution of preferential flow with soil depth varied among slope positions. In HF and MF, macropore flow dominated the A horizon with limited lateral diffusion. However, in the subsoil, although the SAR of all of the stained areas declined, the LWM (quantified as the SAR of yellow and green patches that have a lower concentration of the dye tracer) intensified. In the sandy soils at the LF site, macropore flow via soil fissures was the major type of preferential flow that showed a limited lateral diffusion. In CL, the degree of preferential flow (mainly as finger flows) decreased with soil depth. Based on the SPW profile, flow patterns were classified along soil depth at each site. The lower degree of preferential flow and the reduced SD in agricultural soils demonstrated the substantial impact of soil management on preferential flow and thereby infiltration. Therefore, current agricultural management exacerbates surface runoff and soil erosion and causes ecological degradation and sediment deposition in the Three Gorges Reservoir Area of China.

Additional keywords: Brilliant Blue, infiltration, land management, lateral water mixing, macropore, subsurface hydrology.

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Introduction

Preferential flow in soils refers to the phenomenon of nonequilibrium flow through a small fraction of the pore space, by which water and solutes are rapidly transported to the deep subsurface or groundwater, thereby bypassing large portions of the soil matrix (Beven and Germann 1982; Jarvis 2007; Guo and Lin 2018). Preferential flow paths are usually networks of soil macropores, soil pipes, and other void spaces formed by soil biota activities, such as animal burrows, living or decayed roots, as well as by erosion cavities and freezing–thawing cycles (Noguchi *et al*. 1999; Sidle *et al*. 2001; Beven and Germann 2013; Laine-Kaulio *et al.* 2014). Despite a limited portion in total soil volume, preferential flow transports a considerable amount of subsurface runoff, reaching 34–99% after big storms or irrigations (Tsuboyama *et al.* 1994; Lin *et al.* 1996; Sidle *et al.* 2000; Nieber and Sidle 2010; Sanders *et al.* 2012). Both the vertical percolation and lateral flow along the slope inclination are influenced by preferential flow (Lavee and Poesen 1991; Noguchi *et al.* 1999; Laine-Kaulio *et al.* 2014; Nyquist *et al.* 2018). Therefore, preferential flow has been recognised as an important component of the water flux through the Critical Zone (Jarvis *et al.* 2007; Guo and Lin 2016).

Many laboratory and field techniques have been established to characterise preferential flow at different time-space scales (Allaire *et al.* 2009; Jarvis *et al.* 2016), such as hydrogeochemical monitoring, dye tracing, breakthrough curve method, soil moisture sensor network, and geophysical scanning (e.g. Guo *et al.* 2014; Liu *et al*. 2014; Liu and Lin 2015; Weiler 2017; Guo *et al.* 2018). Among the different methods, dye tracer tests provide the most compelling evidence of preferential flow distribution in the subsurface, and distinguish preferential flow paths from the soil matrix (e.g. Flury and Flühler 1995; Forrer *et al.* 2000; Weiler and Naef 2003; Gerke *et al.* 2015). For example, Öhrström *et al*. (2002) examined the spatial distribution of preferential flow paths in semiarid Tunisia by coupling rainfall simulation and dye tracer releasing. Bargués Tobella *et al*. (2014) performed rainfall simulations and tracer experiments in an afforested parkland to investigate the influence of roots and termite mounds on preferential flow generation in semiarid Burkina Faso. Berli *et al*. (2004) compared preferential flow patterns in compacted and noncompacted soils in northern Switzerland based on the distribution of dye-stained areas. Niu *et al*. (2008) mapped preferential flow routes using dye solution in the forest soils in Gongga Mountain in south-west China. Recently, Liu *et al*. (2014) investigated the vertical distribution of preferential flow paths along an altitudinal gradient in south-west China using dye tracing. Despite many applications of dye tracer to map preferential flow paths in soils, few efforts have been pursued to characterise the lateral water diffusion from preferential flow paths into the soil matrix.

In past studies using dye tracer to determine preferential flow, the colour images of the dye-stained soil profiles were first converted to black-and-white binary images. Then, the preferential flow paths were identified by separating the stained areas from the unstained soil according to a designed threshold of the grey value of pixels (e.g. Flury *et al*. 1994; Laine-Kaulio *et al.* 2015). However, delimitation of such a threshold value was often challenging (Baveye *et al.* 1998). In addition, the degree of the lateral water exchange between preferential flow paths and the soil matrix was difficult to assess (Weiler and Naef 2003). Flury and Wai (2003) suggested that the concentration distribution of the dye tracer could be used to measure the degree of lateral water mixing of preferential flow.

To address this issue, Forrer *et al*. (2000) and Alaoui and Goetz (2008) developed a new method of investigating preferential flow using dye tracing. After maximising the colour saturation of the stains, they divided the images into four categories: dark blue, light blue, green, and yellow patches. Forrer *et al*. (2000) proposed a linear regression method to estimate the concentration of Brilliant Blue dye from the recorded colours. Alaoui and Goetz (2008) demonstrated that the blue patches effectively recovered the preferential flow paths through macropores, and the green and yellow patches indicated the lateral water seeping into the soil matrix. Through determining the degree of lateral water mixing, Weiler and Flühler (2004) categorised the infiltration process into five flow types according to the stained path width (SPW), including three types of macropore flow (low, mixed, and high interactions with the soil matrix) and two types of matrix flow (heterogeneous or homogeneous flow). Weiler and Flühler (2004) considered that long, narrow stripes of dye-stained areas were related to preferential flow with low interaction with the soil matrix whereas wider flow paths indicated a higher degree of lateral water mixing with the soil matrix. This method has enhanced the dye tracer application in quantifying and understanding preferential flow process in field soils (Allaire *et al.* 2009).

In this study, we aim to test this advanced dye tracing method in the Three Gorges Reservoir Area (TGRA) in China. The TGRA (~79 000 km^2) is located in the subtropical region of south-west China, in the transitional zone between the Qinghai-Tibet Plateau and the Middle-Lower Yangtze Plain. The estimated soil loss per year exceeds 153 million tonnes in the TGRA (Lu *et al.* 2003), where severe soil erosion is primarily triggered by extreme scouring of runoff in the unprotected surface soil after deforestation (Cai *et al.* 2005). Given the efficient transportation of stormwater via preferential flow paths that can alleviate the erosive force of overland flow, it is valuable to investigate preferential flow and its controlling factors in this region (Liu and Du 2013; Liu *et al.* 2014). Furthermore, more than 95% of the land area in the TGRA is dominated by mountains and hills, manifesting a significant change in elevation (>2000 m) and local microclimate. In general, the climate in the basal belt of the mountains is warm and humid, but colder and wetter in the mid-mountain and subalpine zones (Shen *et al.* 2001). Along the climate gradient, land uses and soil types vary considerably, showing different soil permeabilities and preferential flow patterns (Liu *et al.* 2014). Recently, after 30-years of cultivation in the TGRA, the local government has launched afforestation projects to combat soil erosion, which also affect soil properties and the infiltration process (Liu and Du 2013). Although some previous studies compared preferential flow between forestlands and deforested fields in the TGRA (e.g. Wang *et al*. 2010; Cheng *et al*. 2011), none investigated the altitudinal patterns of preferential flow and the degree of lateral water mixing across land uses and soil types.

In this study, three forest sites and a deforested site, each representing one typical soil type along the elevation gradient in the TGRA, were selected to investigate preferential flow. Photographs of dye-stained soil profiles together with field measurements of soil physical properties and root distribution were used to measure and understand preferential flow at the pedon scale. The objectives of this study were three-fold: (i) to compare preferential flow patterns under different vegetation covers and soil types, (ii) to examine the degree of lateral water exchange between preferential flow and the surrounding soil matrix across the study sites, and (iii) to identify the dominant factors that control the occurrence and distribution of preferential flow. We hypothesise that the spatial pattern of preferential flow and its lateral water mixing with the soil matrix vary on different hillslope positions in the TGRA due to the altitudinal changes in vegetation cover and soil type.

Materials and methods

Study area

The field experiment was conducted in the Dalaoling National Forest Park (30°00′13″–31°28′30″N, 100°51′8″–111°39′30″E), located in the transitional zone of the upper to middle reaches of the Yangtze River (Fig. 1). The Three Gorges Dam is 40 km downstream of the study site. The local bedrock is crystalline

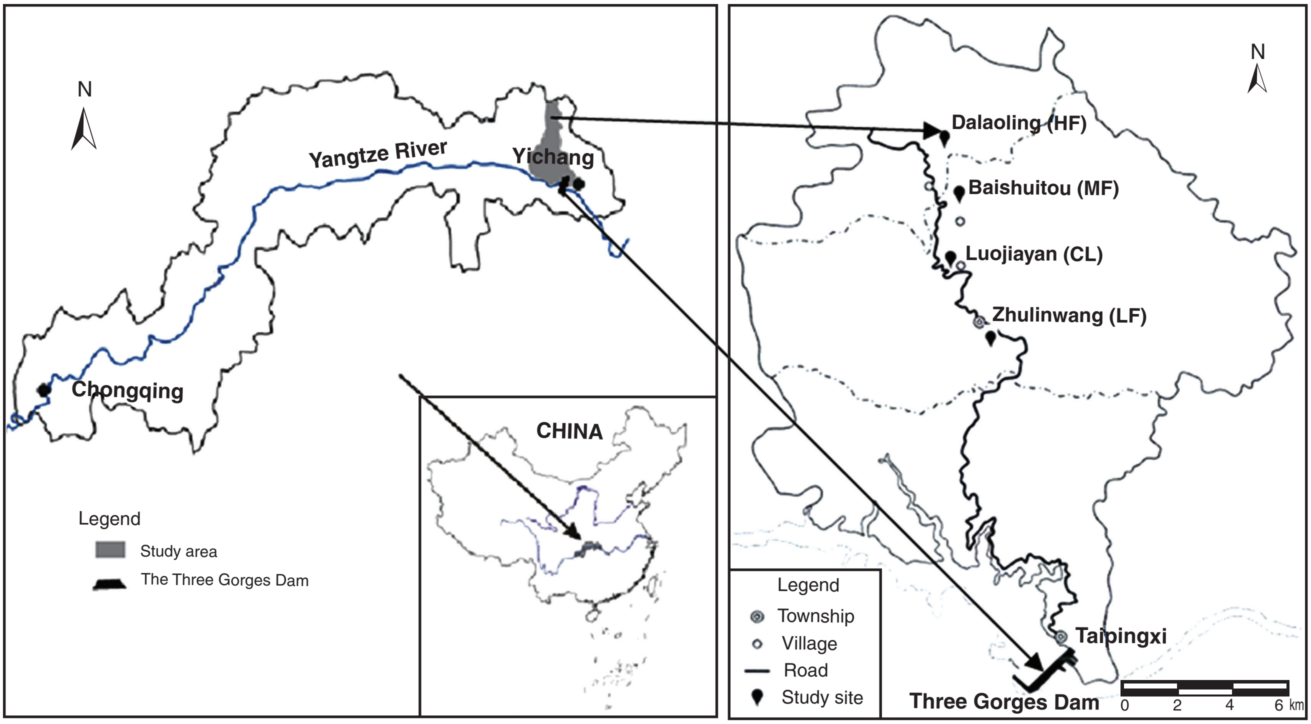

Fig. 1. Study area and locations of the four sites selected for dye tracing tests at different slope positions: HF, high-mountain (subalpine) forest; MF, middle-mountain forest; CL, cultivated land; and LF, low-mountain forest.

rock from the middle–upper Proterozoic erathem base. This area has a subtropical monsoonal humid climate. According to the meteorological records over the period 1994–2014 at a ground station in the study area (200 m a.s.l.), the mean annual precipitation and evaporation are 1011 and 950 mm respectively, mean annual air temperature is 16.7°C, and the mean annual frost-free period is 283 days. Annual precipitation fluctuates within 1000–1340 mm, with the majority occurring as summer storms. Rainfall intensity can be as high as 48 mm/h in July and August (Shi *et al*. 2012), which leads to serious soil erosion and flood hazard. The microclimate shows a vertical variation trend, with temperature decreasing by 0.4–0.6°C and precipitation increasing by 10–15 mm for a 100-m rise in elevation. The climate above the mid-mountain belt is cold, moist, and misty, and the relative humidity exceeds that of the lower belts (Shen *et al*. 2001).

After the long-term tillage and deforestation management, the original evergreen broad-leaved forest dominated by *Castanopsis* spp. and *Phoebe* spp. has undergone considerable changes. The low-mountain forestland (LF) is now covered with secondary shallow-rooted coniferous forests (altitude <800 m a.s.l.), such as *Cunninghamia lanceolata*, *Pinus massoniana*, *Cupressus funebris*, and *Pinus armandii* (Xie and Chen 1998), and the local soils are classified as Udults (Soil Survey Staff 1992). Along with the gradient of elevation and human activity, vegetation cover and soil type show clear vertical zonality. The soils become Ustults in the middle-mountain forest (MF) of 800–1600 m a.s.l., which is covered with broad-leaved coniferous mixed forest, including *Quercus glandulifera*, *Brevipetiolata* spp., *Castanea mollissima*, *Carpinus fargesiana*, *Rhododendron* spp., and *Ilex* spp. In the subalpine forest zone (HF) above 1600 m a.s.l., the typical soil type is Boralfs and is covered with temperate deciduous broad-leaved forest, including *Fagus engleriana*, *Liriodendron tulipifera*, *Castanea mollissima*, *Castanea henryi*, and *Pinus armandii* (Shen *et al.* 2001).

Three forestland sites were selected for preferential flow investigation along the hillslope with the elevation increasing from 700 to 2000 m a.s.l. Additionally, a deforested site in the middle-mountain area was included, which was previously cultivated land planted with sweet potatoes for more than 50 years and has been abandoned since 2010 (Fig. 1). The forest sites included Dalaoling (HF, 1700 m a.s.l.), Baishuitou (MF, 1173 m a.s.l.), and Zhulinwan (LF, 740 m a.s.l.). The deforested site was located at Luojiayan (CL, 942 m a.s.l.), where *Artemisia* sp. was the major vegetation species (~70 plants/m^2). Information on soil properties of each soil horizon for the four study sites is listed in Table 1.

Dye tracer experiments

Three sampling plots were established at each study site to conduct dye tracing experiments, where the local slope gradient was ~25–30°. Brilliant Blue FCF (MedChemExpress, Monmouth Junction, New Jersey, USA) was chosen due to its high visibility in soils, ease of dissolution, and relatively low toxicity to the environment (Flury and Flühler 1995; Weiler and Flühler 2004). A total of 25 L of Brilliant Blue solution (concentration of 4 g/L) was prepared for each site by diluting 100 g of Brilliant Blue FCF powder in 25 L of tap water. The litter layer was carefully removed before dye tracer

Table 1. Soil physical properties, root biomass, and hydraulic conductivity of the study sites

Study sites	Soil taxonomy (USDA)	Soil horizon[A]	Depth (cm)	Bulk density (g/cm^3)	Total porosity (%)	Sand (%)	Silt (%)	Clay (%)	SOC (g/kg)	Root biomass (kg/m^3)	Saturated hydraulic conductivity (mm/min)	Soil water holding capacity (g/cm^3)
Dalaoling (HF)	Boralfs	A	0–15	0.57	56.22	38.08	29.22	32.70	78.94	19.0	8.94	40.2
		E	15–40	0.90	51.08	42.96	37.34	19.70	37.74	6.0	5.41	16.74
		B	40–60	–[B]	–	–	–	–	–	–	–	–
Baishuitou (MF)	Ustults	A	0–15	1.10	48.93	56.28	23.14	20.58	31.38	6.58	8.52	20.48
		E	15–40	1.11	46.96	55.90	23.70	20.40	27.72	6.82	6.04	23.74
		B	40–60	–	–	–	–	–	–	–	–	–
Zhulinwan (LF)	Udults	A	0–7	1.26	26.36	60.04	21.28	18.68	18.35	4.26	0.58	19.65
		C	7–40	1.64	21.5	63.03	17.50	19.47	12.58	3.19	0.2	17.82
Luojiayan (CL)	Arents	A_p	0–15	1.60	36.26	84.48	7.30	8.22	13.56	6.4	0.42	16.62
		B	15–40	1.70	23.50	72.67	11.5	15.83	14.19	2.45	0.24	17.15

[A]A refers to the mineral horizon formed below the O horizon in forest soil and instead by A_p in cultivated land, E refers to the light-coloured mineral horizon below the A horizon with eluviation of decomposed organic material, C refers to the parent material horizon above which mineral soil horizons formed, and B refers to the illuviation horizon where weathering substances were deposited.
[B]No soil sample was collected for the unstained area.

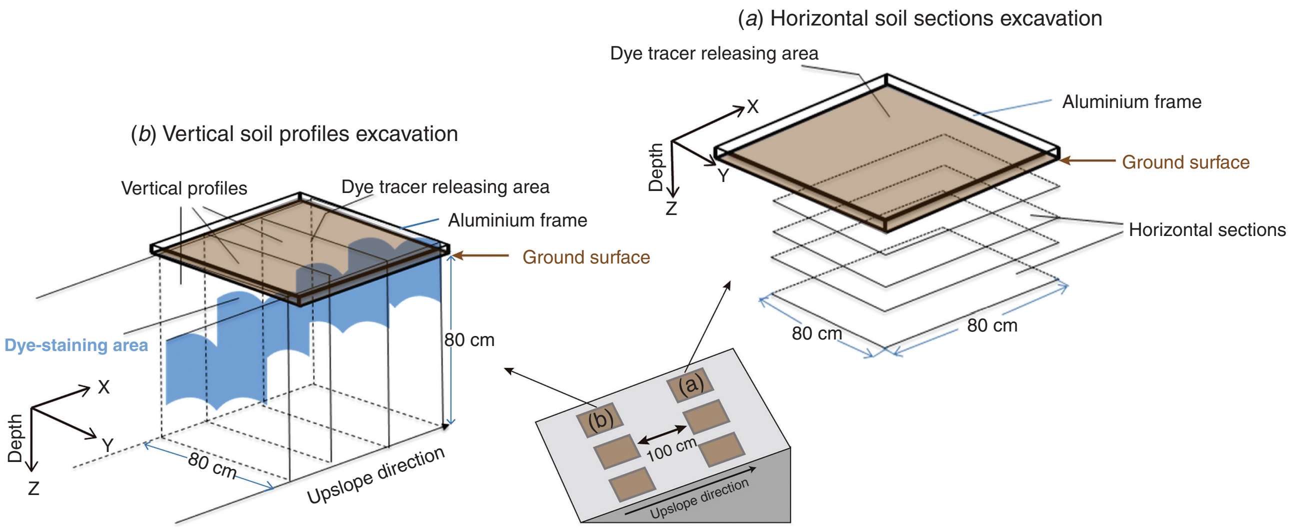

Fig. 2. Scheme of the excavation of dye-staining soil horizontal sections (*a*) and vertical soil profiles (*b*). At each study site, three plots were prepared for dye tracing tests followed by excavation of horizontal soil sections. Three more plots were established 100 cm downslope away to repeat the dye tracing experiments but used for excavation of the vertical soil profiles.

release. The dye solution was sprinkled evenly through a rainfall simulator on the soil surface that was surrounded by a square aluminium frame with an inner width of 80 cm. We adjusted the sprinkling rate according to the infiltration rate to avoid ponded water overflowing from the frame edge. At 24 h after the sprinkling, horizontal soil sections (80 cm × 80 cm) were excavated and trimmed carefully in 5- or 10-cm intervals below the infiltration surface to the bedrock (Fig. 2*a*). Due to the different depths to bedrock, five to seven horizontal soil sections were obtained in each plot. Furthermore, the same sprinkling experiment was repeated 50 cm downslope to excavate parallel vertical soil profiles (Fig. 2). Three to four additional vertical soil profiles with a spacing interval of 10 or 15 cm were obtained, starting at 20 cm away from the frame edge (Fig. 2*b*). Each dye-stained soil profile was photographed with a Pentax KR 848 digital camera (4288 × 2848 pixel sensor resolution) under daylight conditions. A ruler was placed on the soil profiles to assist subsequent image processing and analysis.

Image processing and parameter extractions

To suppress the effect of the frame edge on the infiltration of the dye solution, the central part of the images (60 cm length × 60 cm width for horizontal sections and 60 cm width × 60 cm depth for vertical profiles) were clipped in the image editing program Photoshop CS7 (Adobe Systems Inc., San Jose, California, USA). All horizontal sections obtained at each site were used to quantify the stained area ratio (SAR) and the lateral water mixing between preferential flow paths and surrounding soil matrix (LWM). The vertical profiles were used to determine the deepest stained depth (SD) and the SPW, based on the categorisation of the flow types. Image processing was based on the procedures proposed in Alaoui and Goetz (2008) and Cey and Rudolph (2009), including:

(1) Geometric correction. Because some photographs could not be taken orthogonally in the field, the cutting tool in Photoshop software was used to rectify the warped photographs (Fig. 3*b*) in accordance with the vertical and horizontal scales in the images (Fig. 3*a*).
(2) Separation of the dye-stained areas. Saturation of the blue stains was maximised. This process turned the true-colour image (Fig. 3*b*) into a bright false-colour image (Fig. 3*c*), on which five colours (brown, dark blue, light blue, green, and yellow) appeared. The brown area indicated unstained soil whereas the dark blue, light blue, green, and yellow areas indicated stained sections (Fig. 3*c*). The concentration of Brilliant Blue decreased in the sequence of dark blue, light blue, green, and yellow (Forrer *et al.* 2000). The Magic Wand tool in Photoshop software was used to select and remove the brown area (i.e. the unstained soil matrix), leaving the stained areas in the image (Fig. 3*d*). Next, the remaining four colours (dark blue, light blue, green, and yellow) were selected and clipped using the Magic Wand tool (Fig. 3*e*). The dark blue sections indicated the

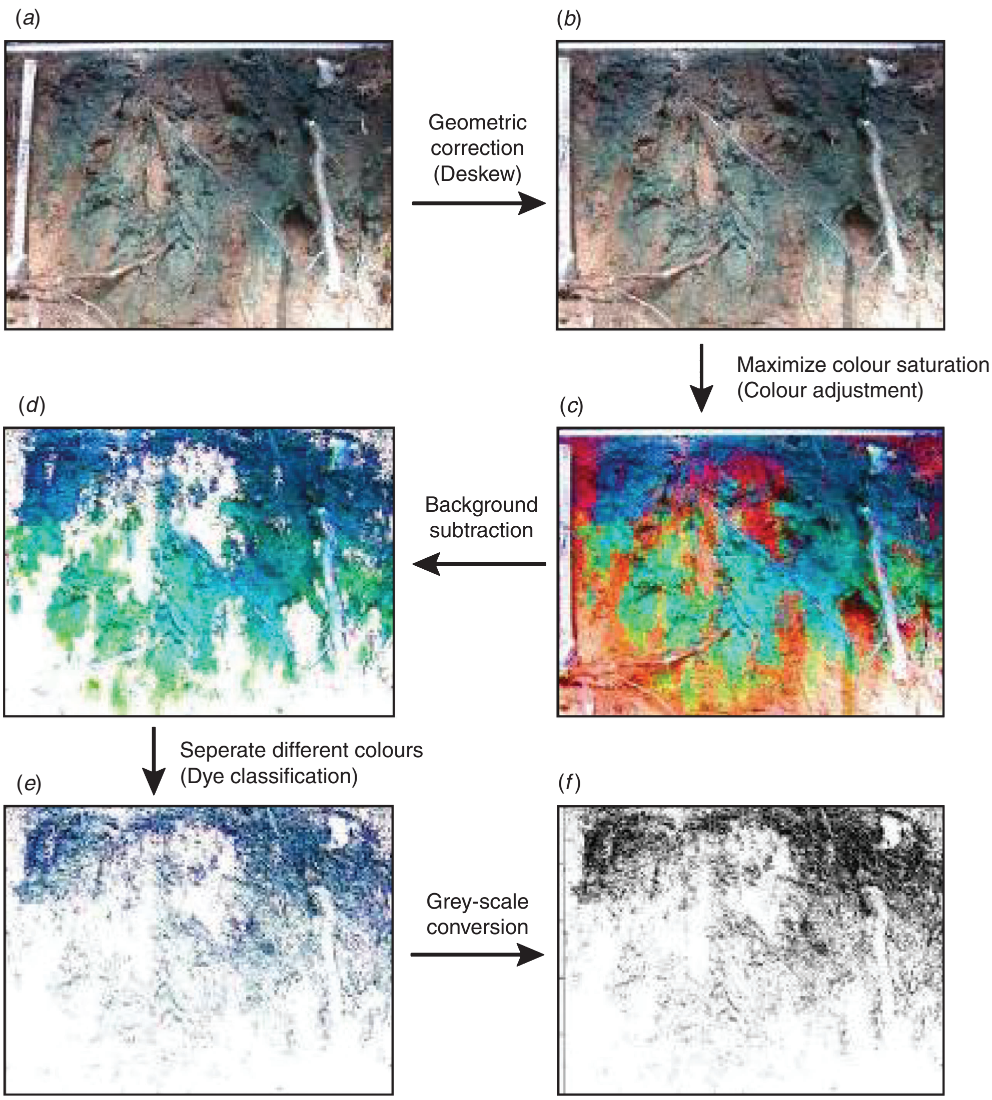

Fig. 3. Flowchart of image processing and parameter extraction. (1) Geometric correction rectifies the warped photographs into an orthogonally view. (2) The true colour image is converted to a false-colour image by maximising the colour saturation, showing five colours (including brown, dark blue, light blue, green, and yellow). (3) The brown area (i.e. the unstained soil background) is removed. (4) The other four coloured areas are separated with dark blue indicating soil pores that preferential flow has passed through, light blue indicating preferential flow path, and green and yellow indicating lateral water mixing. (5) The colour image is then converted a black-white binary image for preferential flow parameter extraction.

empty soil pores that preferential flow had passed through, the light blue sections indicated the preferential flow path (i.e. the wall of the macropores), and the green and yellow sections indicated the lateral water mixing between preferential flow paths and the soil matrix (Alaoui and Goetz 2008).

(3) Parameter extractions in images of horizontal soil sections. After the stained areas of different colours were separated, each colour was exported as a new single-colour image (Fig. 3*e*), which was then converted to a black–white binary image (Fig. 3*f*). The SAR of each colour was calculated using the ratio of the stained pixels to the total number of pixels (Fig. 3*f*). The total SAR of the stained areas was the sum of the SAR of dark blue, light blue, yellow, and green that indicated the overall degree of preferential flow. The LWM was obtained by summing the SAR of the green and yellow patches. The SAR and LWM were quantified at each horizontal soil section to characterise the changes in preferential flow patterns along soil depth.

(4) Parameter extractions in images of vertical soil profiles. The maximum SD was determined based on the depth of the deepest stained patches. The SPW, i.e. the one-dimensional extension of the dye-stained areas at a given depth (also known as the intercept length; Weibel 1979), can be used as a proxy for the size of the preferential flow path. The SPW profile was gained by calculating the frequency distribution of SPWs against soil depth. Then, the SPW was classified into three classes (<20, 20–200, and >200 mm), and the relative frequency of each class to the total width of SPW was calculated. Based on the definition of flow types proposed by Weiler and Flühler (2004), we categorised the stained sections into five flow types: three types of macropore flow (low, mixed, or high interactions with the soil matrix) and two types of matrix flow (heterogeneous or homogeneous flow). Finally, the flow types were compared among the studied soil horizons.

Additional measurements on soil properties and root distribution

Soil horizons were described in excavated soil pits that were ~50 cm away from the dye infiltration area. In addition to the litter layer (O horizon) that was removed before dye release, two to three soil horizons were observed at each site (Table 1) according to the definition of soil horizons described by Schaetzl and Anderson (2005). Disturbed and undisturbed soil samples with four replications in each soil horizon were sampled at all study sites. Disturbed soil samples (500 g each) were indoor air-dried, then ground and sieved to particles with diameter < 2 mm. Particle size distribution was quantified by the pipette method, and soil organic content (SOC) was determined using the potassium dichromate oxidation method of Walkley and Black (1934). Gravel content was measured by weight (Carter and Gregorich 2006). A metal ring knife of 5 cm high and 5 cm diameter was pressed into the soil to collect intact soil cores (100 cm^3 each). Soil bulk density was determined after soil cores were oven-dried at 115°C to a constant weight. Total soil porosity (ϕ_t) was estimated as:

$$\phi_t = (1 - \rho_b/\rho_d) \times 100, \tag{1}$$

where ρ_b is soil dry bulk density and ρ_d is soil particle density (i.e. 2.65 g/cm^3). Root distribution against soil depth was determined by excavating a soil profile and taking soil samples (20 cm × 20 cm × 10 cm) at 10-cm depth intervals to 60 cm deep. The collected soil samples were soaked in water for 24 h. After the roots were washed thoroughly, their diameters were measured by Vernier caliper. Finally, the roots were dried at 60°C to a constant weight to measure biomass.

Results and discussion

Dye coverage in horizontal sections at each study site

Based on the *t*-test, the average SAR of the all stained area significantly ($P < 0.05$) differed among the study sites (Table 2). The largest average SAR was observed in MF (80%), followed by LF (68%), CL (48%), and HF (30%). Additionally, the greatest SD was for LF (69.5 cm), followed by MF (56.4 cm), HF (52.4 cm), and CL (26.5 cm).

The varied SAR values among the forest sites could be attributed to differences in soil texture, SOC, soil structure, and root biomass. For example, HF had the largest SOC (37.74–78.94 g/kg), especially in the 0–15 cm depth layer (78.94 g/kg). Jarvis (2007) showed that more massive organic matter content impeded the development of macropore flow. Guo and Lin (2018) also suggested that surface hydrophobicity, caused by SOC, restrained infiltration and then preferential flow generation in the subsurface. The largest SAR in MF was likely associated with the highest root density (50.1 kg/m^3). Many studies have recognised root channels as an important preferential flow route (Luxmoore 1981; Beven and Germann 1982). Aubertin and Bussiere (2001) reported that due to preferential flow along roots, the permeability of root zone soils of eucalyptus in flood plains was 2–17 times that of grasslands. In addition to SOC and root distribution, soil texture and soil structure also affect the occurrence and dynamics of preferential flow (Guo and Lin 2018). The higher

Table 2. Stained area ratio (SAR) of all of the stained area in horizontal soil sections (60 cm × 60 cm) at different depths of study sites
Different lower case letters indicate a significant difference in SAR at the same depth among study sites at $P < 0.05$

Soil depth (cm)	SAR at different sites (%)[A]			
	HF	MF	CL	LF
5	60^a	95^b	73^a	92^b
10	38^a	89^b	60^c	65^c
15	35^a	90^b	54^c	64^c
20	38^a	83^b	43^a	77^b
30	5^a	73^b	32^c	–[B]
40	2^a	52^b	24^c	62^b
60	0	0	0	50
Mean ± STD[C]	$30^a \pm 22$	$80^b \pm 16$	$48^c \pm 18$	$68^b \pm 13$

[A]HF refers to high-mountain forest; MF, middle-mountain forest; LF, low-mountain forest; and CL, cultivated land.
[B]Note that '–' means no horizontal soil section excavated at this depth.
[C]STD refers to the standard deviation of SAR at different depths at the same site.

sand content in MF and LF than in HF (Table 1) would facilitate occurrence and persistence of preferential flow, consistent with the result of Ritsema and Dekker (2000). The largest SAR and greatest SD in LF were mainly related to a large number of soil fissures at that site.

The SAR in CL (48%) was lower than that in the forest sites of a similar elevation. Soil management-induced soil structure changes, such as the presence of a dense plough pan and the destruction of soil macropores impeded the occurrence of preferential flow (Guo and Lin 2018). Dye testing in Simian Mountain in south-west China also indicated fewer preferential flow paths in farmland than in forestland and shrubland (Cheng *et al.* 2011). In our case, the deforested site (CL) showed a higher bulk density than HF and MF forest sites. This was likely due to the 50-year agricultural management in CL, which indicated fewer pore spaces and limited preferential flow paths (Alaoui and Goetz 2008). Some studies demonstrated that the formation of soil compaction or a dense plough pan after tillage could enhance lateral preferential flow (e.g. Etana *et al.* 2013). However, the focus of our study is vertical preferential flow. Moreover, the lower clay content in CL (Table 1) might also explain the limited magnitude of preferential flow. According to Koestel *et al*. (2012), a threshold proportion of clay content was needed to form stable macropore structures.

Dye coverage at different soil depths

For all the study sites, the SAR of the stained areas changed at different soil depths (Table 2). The largest standard deviation (STD) of SAR values was for HF, indicating the most non-uniform distribution of preferential flow along soil depth. The lowest STD of SAR among depths was for LF, suggesting a relatively stable magnitude of preferential flow at different depths. In general, SAR decreased in deeper soils, and the highest SAR values were at 5 cm depth for all study sites (Table 2). This horizon is usually defined as a mixing zone in solution transport process (e.g. Beven and Young 1988). Dye-stained areas in HF were mainly concentrated at 0–20 cm depth, with less than 5% distributed below 30 cm. In MF and LF, the SAR remained at a high proportion along the vertical soil profile. The SAR in HF was 2% at a depth of 40 cm, which decreased by over 30 times compared with 5 cm depth, whereas the SAR at a depth of 60 cm in LF only decreased by <50%. In CL, SAR decreased linearly with soil depth, and the dye-stained areas were more reduced in relation to MF and LF at the same depth.

The reduced SAR with soil depth was mainly due to decreased porosity in deeper soil. For all study sites, the subsurface layers had a lower total porosity than surface layers (Table 1). Less porosity led to a lower hydraulic conductivity (Table 1), and thereby a decreased degree of preferential flow. Ersahin *et al*. (2002) also reported that the function of macropores in transporting bromide was mainly in the A and B horizons but hampered in the deeper E horizon, as most macropores were created by root growth in the upper soil layers.

The larger SAR in the deep soils in MF and LF could be a result of the well-developed root system and abundant fissures at the two sites respectively. In contrast, the stained area decreased sharply in the deeper soils in HF. The higher SOC (78.94 g/kg) of the surface soil (0–15 cm) in HF restrained the infiltration rate at the surface and the development of preferential flow in the subsurface (Guo and Lin 2018). Moreover, the larger water holding capacity of the upper layer in HF impeded deep percolation through preferential flow paths. The decrease of SAR in LF at depths of 10 and 15 cm could result from soil clogging by the leaching of soil particles according to our field observations. The lower SAR of all soil depths in CL compared with the MF and LF was likely caused by soil management, e.g. tillage, which led to soil compaction and destruction of soil macropores. The fewer preferential flow paths caused by farming lowered its function in soil and water conservation due to reduced infiltration capacity (Liu and Du 2013).

Lateral water mixing of preferential flow and flow type classification

Lateral mixing and flow types in A (or A_p) horizon

In HF, the LWM (quantified as the sum of the SAR of green and yellow) decreased from 40% to 16% from 5 to 15 cm depths in the A horizon (Fig. 4). At the near-surface layer (5 cm depth), HF showed the largest LWM among the study sites and was likely associated with the shallow rooting depth at this site (Fig. 5). *In situ* excavation indicated that the blue sections concentrated on the coarse roots (with diameter of 0.5–3 cm) whereas the green and yellow sections diffused outside blue sections (Fig. 6), suggesting lateral water mixing in the rhizosphere. Macropore flow with limited interaction with the soil matrix was the main preferential flow type at this site, except for a little mixed interaction at the surface layer (Fig. 7).

In MF, a relatively lower LWM value was observed in the A horizon (Figs 4 and 5). The flow pattern was mainly characterised by the mixed interaction and more permeation (Fig. 7). The soil in the MF had abundant fine roots (diameter < 0.5 cm) but few coarse roots (Fig. 6). Compared with HF in which coarse roots were more abundant, the degree of lateral water diffusion was lower from the fine root channels, likely due to a greater capillary effect (Fig. 5).

The LWM in LF reached 37% in the A horizon (Figs 4 and 5). This implied that lateral diffusion was an important form of subsurface water transport at this site, showing the mixed interaction and heterogeneous flow types (Fig. 7). Due to a thinner A horizon and fewer roots (9.3 kg/m^3; also see Fig. 6), a more substantial amount of dye tracer could be held in the upper soil layers for an extended period, which allowed for the sufficient lateral diffusion to the soil matrix (Fig. 5).

In CL, the LWM remained at a low value in the A_p horizon (0–15 cm depth; Figs 4 and 5). The water flow patterns were classified into heterogeneous and homogeneous flows (Fig. 7), suggesting fewer macropores and a limited degree of preferential flow in CL. Soil macropores in CL were destroyed by previous agriculture cultivation, leading to less porosity in the plough layer (Zhang *et al.* 2014). Moreover, finger flow occurred at this site that bypassed the bulk of the soil (Fig. 6). This phenomenon was also reported by Zhang and Wang (2008), and showed that heterogeneity of soil texture and

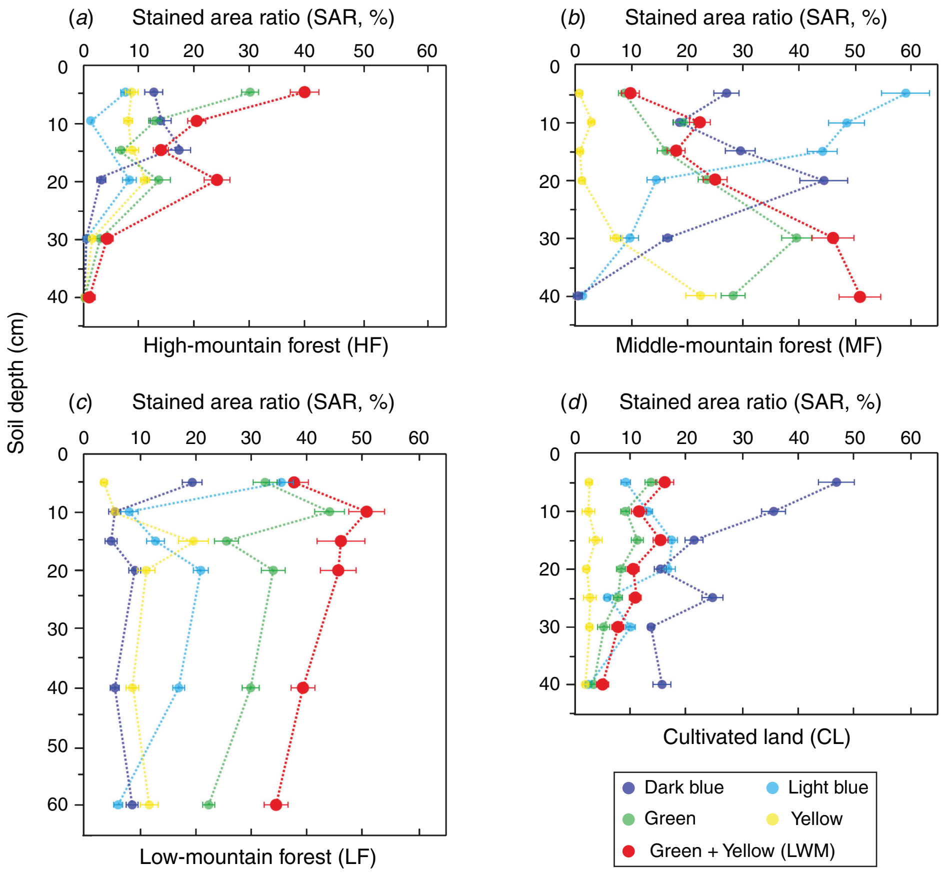

Fig. 4. Stained area ratio (SAR) of different coloured patches with soil depth at different study sites. The SAR of the dark blue and light blue patches indicates preferential flow paths. The SAR of the green and yellow patches (indicated by red dots) refers to the degree of lateral water mixing (LWM) from preferential flow paths into the soil matrix. Error bars indicate the standard deviation among the three replicated horizontal soil sections at each site.

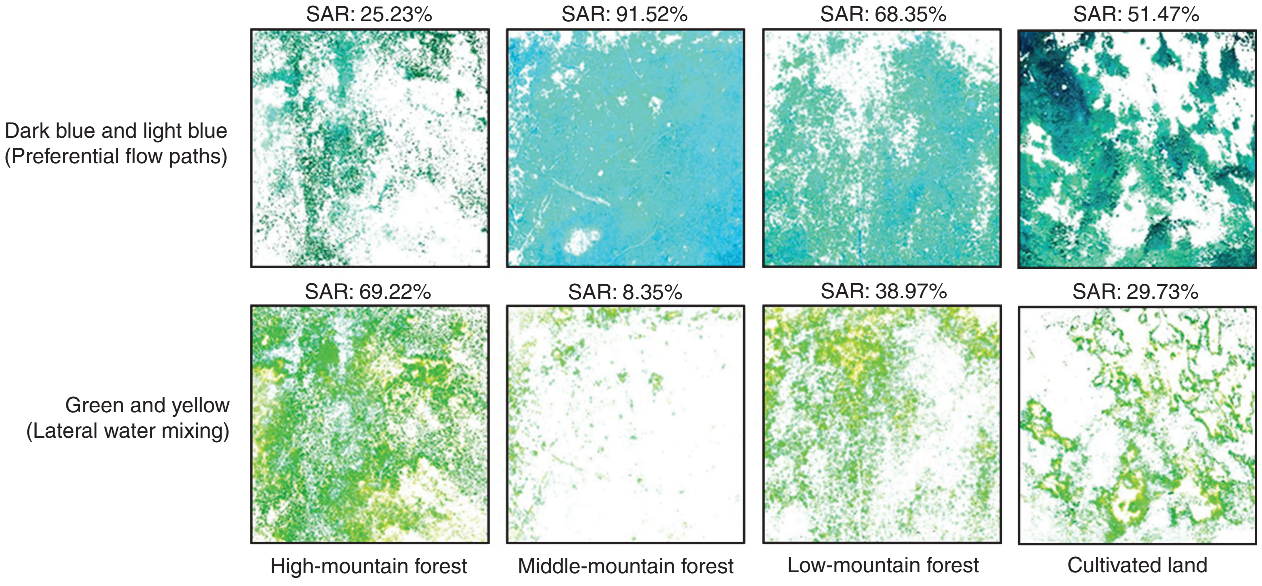

Fig. 5. Comparison of dye-stained patterns in the shallow horizontal soil sections at 5 cm depth among the four sites. Upper row: dark blue and light blue patches, indicating preferential flow paths. Lower row: green and yellow patches, indicating the lateral water seeping from preferential flow into the soil matrix.

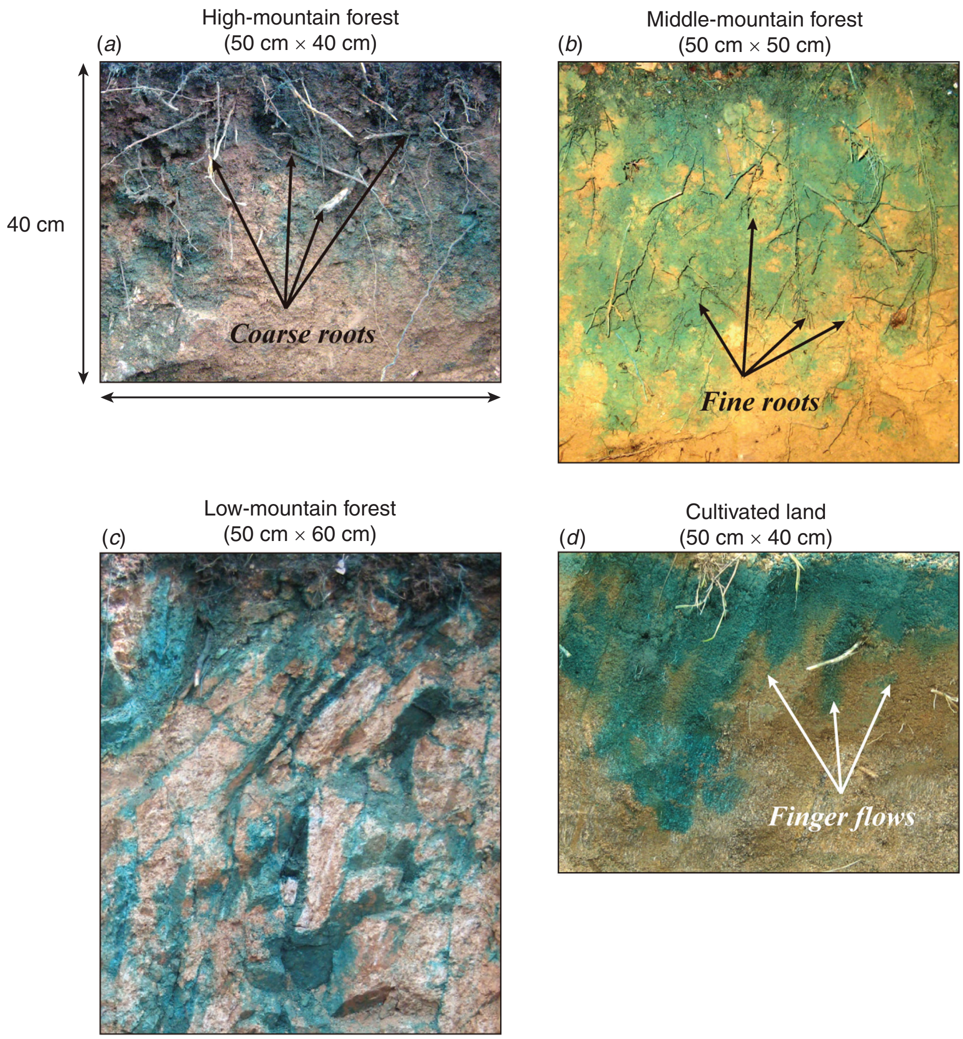

Fig. 6. Examples of the vertical dye-stained soil profiles at the four study sites.

difference in permeability between the upper plough layer and bottom compacted horizon controlled the occurrence of finger flow.

Lateral mixing and flow types in the E (or B, C) horizon

The SAR of the blue patches at a depth of 30 cm in the HF decreased to 0.3%, suggesting that preferential flow was nearly absent below the E horizon (Figs 4 and 8). However, the LWM showed a different pattern: expanding at a depth of 20 cm, but dropping to 5% at 30 cm. The relatively shallow SD (52.4 cm at this site) and the more significant LWM indicates the development of subsurface lateral flow along slope gradient when the percolating encountered less permeable layers (e.g. Laine-Kaulio *et al.* 2015). Thus, subsurface lateral flow or biomat flow likely occurred in HF during heavy rainfall. In another forested hillslope, Sidle *et al.* (2007) detected subsurface lateral flow at the A–E interface and biomat flow within the litter layer.

In MF, the LWM increased with increasing depth continuously in the E horizon within 10–40 cm depth (Figs 4 and 8). The flow regime was dominated by macropore flow with the mixing interaction with soil matrix for 10–33 cm depth, and by macropore flow with less lateral water exchange in the deeper soils (Fig. 7). The variation of stained patches with depth was mainly attributed to the root distribution profile (Fig. 6). The intensive root system distributed in the depth of 0–30 cm could sustain preferential flow. In contrast, less roots in the shallow soils led to the lower SAR of blue patches where the preferential flow process was replaced by lateral water diffusion, as more yellow and green patches became evident (Figs 4 and 8). The LWM behaviour tended to intensify with soil depth due to fewer coarse roots and a higher bulk density in deeper soil (Alaoui and Goetz 2008).

In LF, the LWM remained at a relatively high degree, reaching 40–56% in the subsoil (20–60 cm depth), suggesting extensive water exchange between macropores and the soil matrix (Fig. 4). The flow patterns changed sharply along the vertical profile, possibly a result of heterogeneity of soil structures (Fig. 6). The gravel content (>2 mm) reached 50% in the subsoil, which could facilitate the formation of relatively large voids between gravel particles and trigger preferential flow

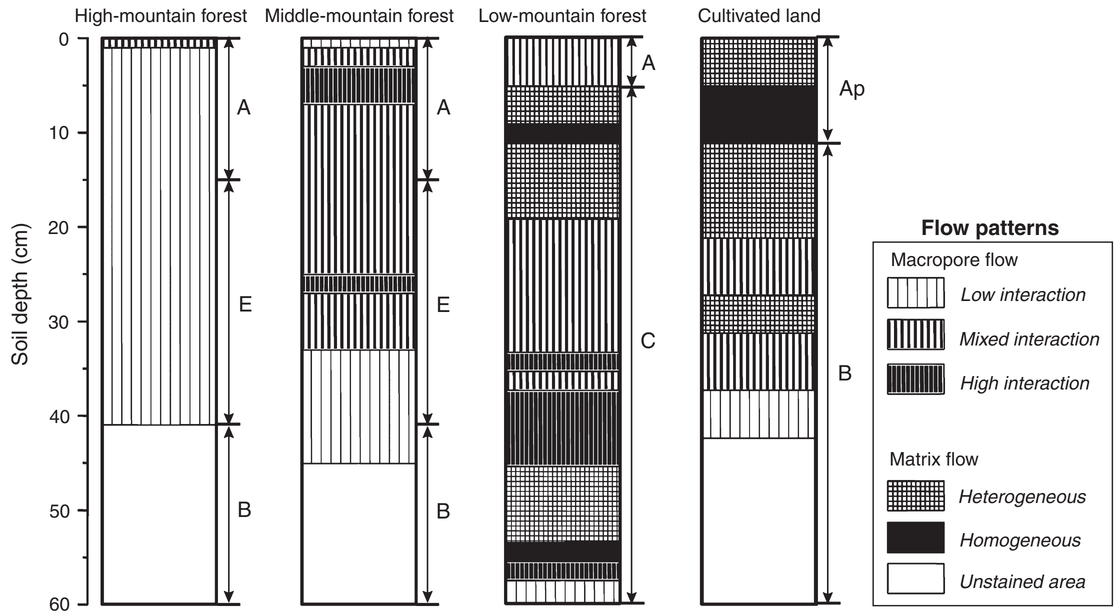

Fig. 7. Flow types along the soil profile at the four study sites. Flow type is identified based on the stained path width (SPW). Following Weiler and Flühler (2004), we categorized five flow types, including three types of macropore flow and two types of matrix flow.

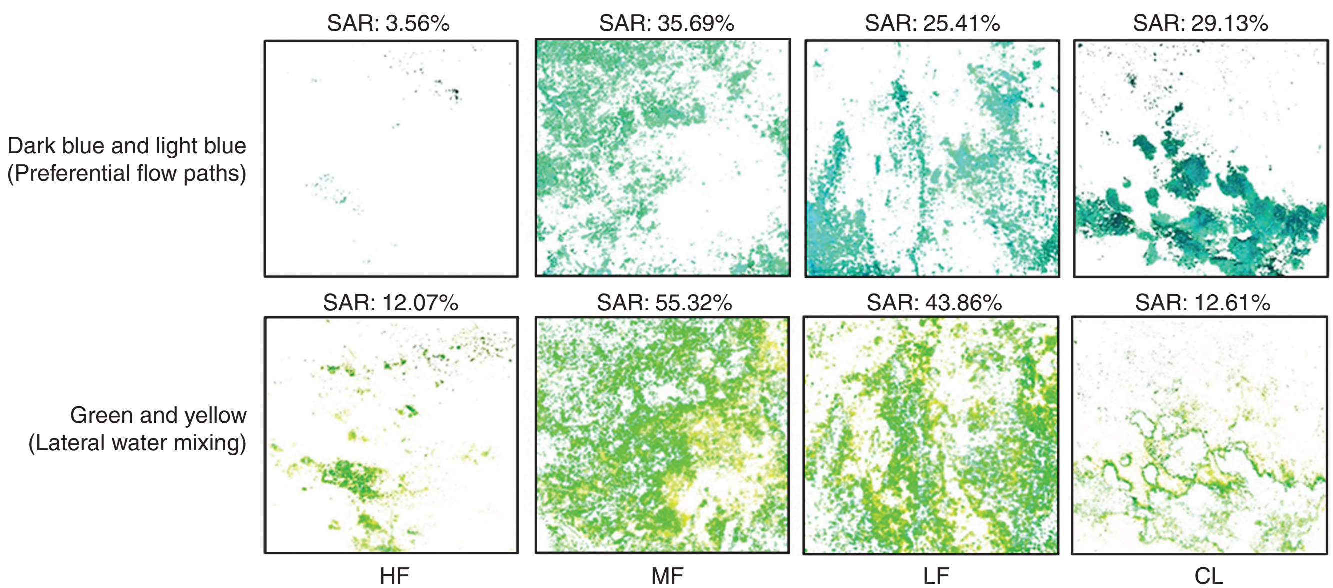

Fig. 8. Comparison of dye-stained patterns in the deeper horizontal soil sections obtained in HF (high-mountain forest; 30 cm depth), MF (middle-mountain forest; 30 cm depth), LF (low-mountain forest; 40 cm depth), and CL (cultivated land; 30 cm depth). Upper row: dark blue and light blue patches, indicating preferential flow paths. Lower row: green and yellow patches, indicating lateral water seeping from preferential flow into the soil matrix.

(Ma 2007). Preferential flow through cracks in the weathered bedrock was revealed by the blue stained sections at a depth of 40 cm (Fig. 6). The vertical distribution of tracer was categorised as macropore flow with low or mixing interactions (Fig. 7). Flow along unsaturated cracks in weathered bedrock is a common preferential flow type in granite mountain soils (e.g. Wang *et al.* 2010; Li *et al.* 2011).

The LWM of the upper B-horizon in CL decreased uniformly with increasing depth (Fig. 4). The CL was covered by wormwood with shallow and fine roots, which was not favourable for preferential flow generation. The preferential flow occurred as finger flows (Fig. 6) due to the significant difference in soil properties and permeability between the upper plough layer and the underlying compacted horizon (Zhang and

Wang 2008). In addition, the SAR of blue patches at a depth of 30 cm was much smaller than that at 5 cm (Figs 4, 5, and 8) because the bottom compacted horizon had a higher bulk density and thereby a lower porosity (Table 1). These findings indicate that deforestation and ploughing management in CL have led to a considerable decrease in infiltration capacity (Liu and Du 2013). As a result, the risk of soil erosion increased significantly, which is the main ecological risk in the investigated region.

Conclusions

Dye tracing experiments and image analysis determined preferential flow paths and their lateral diffusion into the surrounding soil matrix along an altitude gradient in forested and agricultural soils in the TGRA in China. The larger SAR in the MF (80%) and LF (68%) were likely associated with the abundant root distribution and coarse soil texture respectively. The lower SAR in the HF (30%) was mainly caused by the shallower rooting depth and the higher content of clay and soil organic matter. The lower SAR in the CL (48%) was affected by agriculture activity that destroyed macropores and increased soil bulk density. Preferential flow patterns varied among the study sites: macropore flow with low interaction was the primary preferential flow type in the HF, and macropore flow with mixed interaction dominated soils in the MF. Macropore flow and matrix flow dominated the flow patterns in the LF and CL. Limited lateral water mixing from preferential flow paths to the soil matrix was observed at the 0–20 cm depth in the HF due to the root distribution profile, whereas the abundant fine roots in the MF induced a higher degree of lateral mixing. These findings improve understanding of the occurrence and dynamics of preferential flow in mountain areas with different vegetation types, soil properties, and degrees of human perturbation. This knowledge should provide a constructive basis for water resource management, soil water conservation, and vegetation recovery, given that serious soil erosion is the major ecological risk in the study region.

Conflicts of interest

The authors declare no conflicts of interest.

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